

How Testing Of IoT SYSTEMS Differs

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An overview of upgrades in test and measurement devices to meet the requirements of IoT systems



Nowadays most electronic devices are being designed with a touch of the Internet of Things (IoT). Test and measurement (T&M) plays a big role in successful mass-deployment of the IoT. Expanding applications and newer protocols require T&M devices for IoT applications to upgrade simultaneously. What are these upgrades and how are they changing the paradigm of T&M for the IoT? This article answers such questions.

Testing an IoT system

Optimal testing of an IoT system depends on how its intricate components are tested and measured. The IoT encompasses different wireless technologies like Wi-Fi, Bluetooth, cellular (2G, 3G, LTE and narrow-band IoT), Zigbee, LoRa and so on. It draws strength from the level of connectivity these technologies provide.

Mombasawala Mohmedsaed, general manager (applications), Keysight Technologies India, explains, "Test gears are available to measure the functioning of these technologies overall. However, the importance lies in having a low-cost T&M solution for testing the individual IoT devices involved in these technologies. The devices themselves are low-value instruments capable of connecting to the network and sending or receiving commands."

So what are the essential layers that

an engineer needs to test in order to ensure an efficient IoT system? Mombasawala answers, "Engineers mainly look at transmitter and receiver measurements, which are radio-frequency (RF)-based. Depending upon the chosen wireless technology, manufacturers perform tests like spectral emission measurement, channel bandwidth measurement, receiver sensitivity measurement and so on. All these measurements of an RF device are based on power, frequency and bandwidth. Another critical measurement required in the IoT is battery drain analysis. There are certain test gears in which the current and voltage cycles can be varied and one can bleed out the battery to test how long it can support a real-time application and how long it lasts (battery life). With the right chipset, a strong battery and the small unit of a transceiver, the IoT device is ready to connect to the network."

T&M devices

Test equipment manufacturers focus on setting up appropriate firmware to support the different IoT bandwidths and frequencies. They are removing the redundant hardware from test devices and bringing down the cost of test devices in proportion to the scale of design.

Mombasawala talks about Keysight's equipment roster as an example, "IoT chipset designers need a full-blown testing system that will support 2G, 3G, 4G and